Fiscal Year:	FY 2025	**Task Last Updated:**	FY 11/05/2025
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Project Title:	Figures of Merit for Space-Based Production of a Therapeutic in Plants		
Division Name:	Human Research		
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Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	95616-5270	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2024 TRISH FOM BEES 2024
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX16AO69A-FMBS0101		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	We propose to conduct a quantitative Figures of Merit (FOM) analysis to evaluate the “cost” of a new innovative capability for biomanufacturing protein-based medications using plants for deep space missions. We will use experimental process methods/data which we have previously generated on plant-based production of a biologic to maintain bone mass density, parathyroid hormone Fc fusion protein.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	There are several benefits of this project. First it has provided a general framework and step-by-step procedure for doing a figure of merit (FOM) analysis that could be used for any production process whether it be for a therapeutic (as in our case), chemical, material, food, enzyme, etc. This will benefit the space community as they consider different ways to make needed compounds on the moon, Mars, or in transit. But this framework and the resulting FOM metrics can also be valuable for comparing alternative production strategies on Earth, particularly in resource limited environments such as remote areas or battlefields where constraints on system size, volume, supply chains, power, personnel/operator time, and waste are important considerations. Second, the FOM analysis can help identify specific challenges for a given production scenario, opening up new R&D efforts. For example, in one of the therapeutic production scenarios we considered, the crew time FOM was impractically high (CM-hrs per day exceeded the number of available hours in the day), indicating that automation of a key step would be required to make that production system feasible. Third, because we allow the weekly dose levels for the therapeutic and the production levels in the plant system to vary in the model, it could easily be adapted to different therapeutics (requiring different doses) and/or different plant hosts and/or gene delivery efficiencies. Finally, we proposed a novel Equivalent System Mass (ESM) metric that utilizes the FOM data , in conjunction with appropriate equivalency factors, to combine the FOM into a single metric that can be used to compare alternative production scenarios. It is novel in the sense that it provides "credits" for co-production of useful outputs that otherwise would have to be flown or produced separately.
Task Progress:	We conducted a quantitative Figures of Merit (FOM) analysis to evaluate the “cost” of a new innovative capability for biomanufacturing protein-based medications using plants for deep space missions. We used experimental process methods/data which we have previously generated on plant-based production of a biologic to maintain bone mass density, parathyroid hormone Fc fusion protein (PTH-Fc), as well as literature data. The FOM metrics we determined were mass (kg), volume (m^3), power (kW), and crew time (crew member (CM) hours). We considered six different plant-based production platforms: transgenic lettuce (Lactuca sativa) producing PTH-Fc, transient production of PTH-Fc in lettuce using the gene gun for PTH-Fc gene delivery to the plant, transient production of PTH-Fc in Nicotiana benthamiana using vacuum agroinfiltration for gene delivery, transient production of PTH-Fc in Nicotiana benthamiana using a engineered plant virion for gene delivery, transgenic Nicotiana benthamiana producing PTH-Fc and flown PTH-Fc in injectable syringes. For each system we constructed block flow diagrams identifying system components, inputs, waste outputs and useful outputs, and calculated the mass (kg), volume (m^3), power (kW), and crew time (CM-hr) for a 600 day bioproduction period on the surface of Mars, for different target weekly doses of PTH-Fc, per CM over the 600 day period. Only the upstream (production of PTH-Fc) part of the process was included in this analysis since the downstream (purification of PTH-Fc from the plant material) may not be needed if the drug can be delivered to the CM orally, and since data were not readily available for downstream processing steps. Key findings were that the product titer (how much PTH-Fc could be produced per fresh weight of plant tissue), weekly dose needed to be produced, and refrigeration requirements had a significant impact on the FOM. In some cases, crew time was impractically high, indicating the need for automation in the production process. In general, the transgenic PTH-Fc lettuce, transgenic PTH-Fc Nicotiana benthamiana and transient production using engineered plant virus in Nicotiana benthamiana had the lowest FOM under the model assumptions used. The impact of this work is that the general framework and workflow can be used to develop similar models for other bioproduction systems. For the PTH-Fc study, the model allows users to determine FOM for different plant-based PTH-Fc production scenarios, and those results can be used to determine if a particular production scenario would meet specific space mission constraints associated with mass, power, volume or crew time, or under what conditions (e.g. PTH-Fc titer) it could meet those constraints (indicating targets for further process development). In addition this type of analysis would be valuable for deploying bioproduction systems in other resource-limited environments, such as remote areas on Earth or in battlefields. Although this is our final report, we plan to work on a publication of our findings, and in the future, apply FOM analyses to other bioproduction processes to assess their "cost" and appropriateness to different resource constrained operating environments.
Bibliography Type:	Description: (Last Updated: 11/05/2025)
Awards	Haddad K. "Business Development Fellowship." Aug-2024
Awards	Haddad K. "Innovation Institute for Food and Health Summer Entrepreneurial Fellowship." Jun-2025
Awards	Haddad K. "UC Davis Graduate Students Association Travel Award." Mar-2025
Awards	Taylor S. "Business Development Fellowship." Aug-2024